
Specialist Physical AI Need Not Be Opaque

A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control, failure analysis, and weight-derived causal structure

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Tensor decomposition turns a large table of learned interactions into simpler low-
2 rank factors that can be ranked and inspected. Conventional robot policies con-
3 tain operations that block this direct weight-level view. We introduce χ -VLA, a
4 20.1M-parameter vision-language-action policy built only from bilinear, linear, and
5 rational learned operations. Its weights can therefore be reorganized into explicit
6 interaction tensors rather than treated only as an opaque checkpoint. A mechanical
7 audit and numerical reconstructions verify that the deployed policy preserves this
8 form. Trained from scratch on LIBERO-Object, convolutional χ -VLA reaches
9 $93.7\% \pm 0.8\%$ over three seeds under the published protocol. Exact weight-derived
10 attention subspaces beat equal-rank random controls in all 36 primary block-head
11 tests. Retaining 64 of 384 causally ranked visual dimensions preserves 18/20
12 successes, while a random subspace preserves none. Yet an 80-trial instruction in-
13 tervention exposes a scene-identity shortcut, and the first direct coefficient edit fails
14 its selectivity gate. The result is a capable and structurally inspectable specialist,
15 not a zero-shot generalist or a solved editing interface.

16 1 Introduction

17 Modern robot policies can achieve high success while giving little indication of why an action was
18 chosen. Their checkpoints contain millions of weights, but those weights do not directly show which
19 image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated
20 activations, yet a correlation is not automatically a mechanism or a safe editing target.

21 A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of
22 low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses
23 additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction
24 table defined by its weights. Its decomposed factors then provide candidate control mechanisms that
25 can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully
26 tensor-decomposable policy.

27 The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization
28 break the algebraic form, while continuous actions require precise closed-loop outputs. The potential
29 payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows
30 language, keys on a visual shortcut, or uses a fragile control feature.

31 We study the narrower but testable question:

32 *Can a specialist VLA retain competitive closed-loop capability when every deployed*
33 *learned operation is polynomial or rational, and does the resulting structure*
34 *provide a useful weight-based analysis surface?*

35 The stronger thesis asks for three results on the same policies: small capability cost, stable weight-
36 derived causal structure, and selective edits without retraining. We establish the first two only partially
37 and report the failed first surgery screen rather than treating algebraic representability as proof of
38 interpretability.

39 Our contributions are:

- 40 1. We construct a complete VLA from tensor-compatible primitives. A rational normalization
41 supplies input-specific scaling without leaving the rational function class.
- 42 2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with
43 published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$
44 over three seeds, and a three-checkpoint ensemble reaches 96.2%.
- 45 3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational
46 and bilinear branches from their weights.
- 47 4. We extend exact weight-derived decomposition through individual attention layers with a
48 reduced-Gram calculation that avoids materializing the full high-order tensor.
- 49 5. We show a causal closed-loop consequence. A selected 64-dimensional visual subspace
50 preserves almost all tested behavior, while a random equal-width subspace fails. A separate
51 instruction intervention exposes a real scene-identity shortcut and limits the language-
52 grounding claim.

53 This paper does not claim current state of the art. It also does not claim that comparable mechanisms
54 are impossible to obtain from conventional policies. The empirical question is whether the algebraic
55 policy makes them more direct, stable, and selectively editable than matched post-hoc methods.

56 2 A rational tensor robot policy

57 2.1 Tensor-compatible primitives

58 Each primitive replaces a familiar transformer component while keeping explicit coefficients. Lin-
59 ear maps contribute first-order terms, while multiplying learned projections creates higher-order
60 interactions that remain determined by the weights.

61 **Homogeneous coordinate.** We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative
62 layer to represent bias, linear, and quadratic terms in one contraction.

63 **Bilinear feed-forward block.** For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

64 The inner sum is a table indexed by one output and two input coordinates. The three matrices give its
65 CP factorization, the tensor analogue of a low-rank matrix factorization.

66 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

67 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
68 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
69 interaction coefficients determined by the weights.

70 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 71 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

72 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
 73 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
 74 rescaling without leaving the rational tensor representation.

75 2.2 Architecture and training

76 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
 77 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
 78 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
 79 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
 80 continuous action dimensions. The complete model has 20,137,352 learned parameters.

81 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 82 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 83 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 84 that verify every requested state was loaded.

85 2.3 Mechanical certificate on the trained checkpoint

86 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
 87 point at runtime and reconstruct representative operations from saved weights.

88 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 89 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 90 module’s float32 statistic. The audit certifies operator membership and local reconstruction, as
 91 summarized in Appendix Figure 6. It does not materialize one compact polynomial for the full
 92 eight-layer transformer.

93 3 Protocol-aligned closed-loop capability

94 3.1 Evaluation protocol

95 We evaluate all ten LIBERO-Object tasks with:

- 96 • 50 canonical trials per task
- 97 • a 280-step cap
- 98 • three independently trained seeds per architecture
- 99 • 500 rollouts per seed and 1,500 rollouts per architecture

100 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
 101 rerun in our codebase, so differences may still include implementation and training choices.

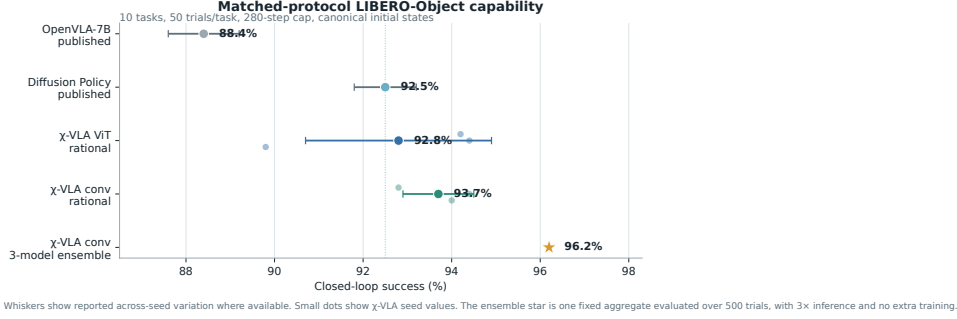


Figure 1: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational ViT	20.1M	3	$92.8\% \pm 2.1\%$
χ -VLA, rational conv	20.1M	3	$93.7\% \pm 0.8\%$
χ -VLA, conv ensemble	$3 \times 20.1M$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The ViT seeds score 89.8%, 94.4%, and 94.2%. The convolutional seeds score 94.0%, 94.4%, and 92.8%. The convolutional model is modestly higher and more consistent in this three-seed comparison.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It does not isolate the cost of decomposability. A decisive cost experiment must train a same-skeleton conventional transformer and staged partial-conversion controls using identical data, updates, seeds, and evaluation states. We treat that experiment as required before restoring the stronger title claim that convertibility is nearly free.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit

numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16} . Agreement with the real trained module ranges from 3.10×10^{-7} to 1.03×10^{-5} because the deployed implementation intentionally casts the statistic to float32.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors rank important directions without constructing the full table. It matches brute-force materialization to 2.13×10^{-14} at toy scale.

4.2 Trained policy result

We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached activations are used only afterward to score how well it preserves the layer’s computation.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Causal policy structure and a language shortcut

We next test whether the identified structure affects closed-loop behavior and whether high task success reflects robust use of the language input.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we rank directions separately for translation, rotation, and gripper actions. A subspace is a smaller coordinate system inside the model’s 384-dimensional visual representation. Retaining it means zeroing all other directions, then running the actual policy. We compare selected and random equal-width subspaces. Because the ranking uses held-out activations and downstream sensitivity, it tests a causal bottleneck but is not a weight-only discovery result.

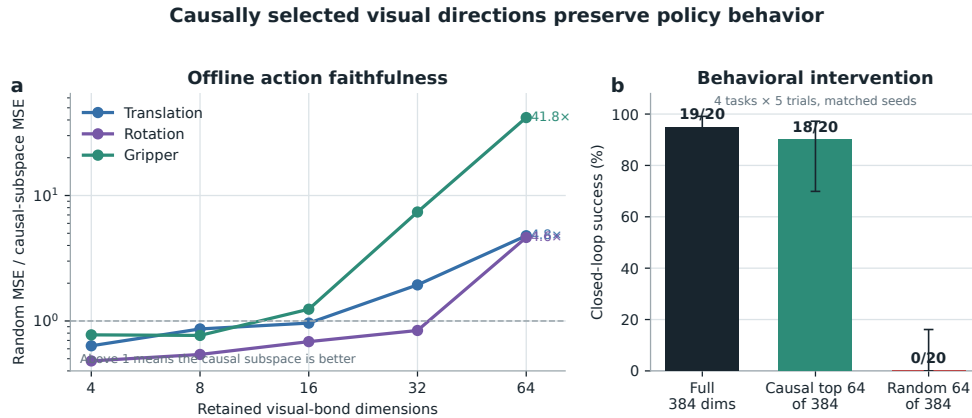


Figure 2: **Causally selected visual directions preserve policy behavior.** The left panel reports random MSE divided by selected-subspace MSE. The right panel intervenes on the real policy over four tasks and five paired trials per task.

155 With 64 of 384 dimensions retained, the selected subspace reduces action reconstruction error relative
156 to random by $4.8\times$ for translation, $4.6\times$ for rotation, and $41.8\times$ for the gripper. In closed loop, the
157 full representation succeeds in 19/20 trials, the selected subspace in 18/20, and a random subspace
158 in 0/20.

159 This result establishes a causal low-dimensional substrate, but it is still limited to one checkpoint and
160 twenty rollouts. The current ranking at this full-policy bond uses held-out examples and downstream
161 sensitivity. The exact layerwise attention method in Section 4 must still be connected to the same
162 full-suite closed-loop intervention.

163 5.2 Counterfactual instruction test

164 High success does not guarantee robust language use. We hold the image and robot state fixed
165 and replace the instruction with one naming another visible object. Across 80 canonical one-step
166 trials, the predicted reach direction shifts toward the newly named object in only 15.0%. Mean
167 directional cosine to the named object falls from 0.525 under the true instruction to 0.299 under the
168 counterfactual instruction, while cosine to the original target remains 0.464. In an earlier 40-step
169 closed-loop screen, 33.3% finish closer to the new object and 55.6% move measurably toward it.

170 The likely shortcut is structural in the dataset. Each scene template is paired with an almost fixed target,
171 so scene identity can substitute for object-name grounding. Decorrelated scripted demonstrations
172 improve the counterfactual metric, but all three fine-tuning attempts severely reduce ordinary closed-
173 loop success. The best repaired dataset raises capability only to 21.5%, far below the healthy
174 range.

175 This negative result narrows the VLA claim. The current model is a capable language-conditioned
176 specialist, but it is not yet a robust instruction-grounded policy. LIBERO-CF independently documents
177 the same broad failure class in existing VLAs [10]. Our prospective novelty is therefore not the
178 shortcut alone. It is the harder test of whether exact weight-derived terms can localize and selectively
179 repair that shortcut.

180 6 Related work and limitations

181 **Related work.** OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control
182 baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative
183 weights can expose low-rank structure and causal controls [3–5]. Recent work also steers pretrained
184 VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a
185 policy designed as a rational tensor program. Tensor and polynomial networks provide the broader
186 multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success
187 can coexist with weak language use [9–11].

188 **Limitations and conclusion.** No same-skeleton conventional twin yet measures conversion cost.
189 The benchmark is one specialist suite, the policy fails a counterfactual instruction test, and exact
190 decomposition is layerwise rather than end to end. The closed-loop subspace test uses one checkpoint
191 and twenty trials, while the first direct coefficient edit fails its selectivity gate. The defensible result
192 is therefore narrow: rational restrictions do not prevent competitive LIBERO-Object control, and
193 they provide a precise weight-derived analysis surface. Broader grounding, matched controls, and
194 validated coefficient edits remain necessary.

195 **Reproducibility.** The final artifact will provide code, configurations, checkpoint hashes, exact
196 certificates, and per-episode logs. Canonical-state checks fail loudly when requested states are
197 unavailable.

References

- [1] M. J. Kim et al. *OpenVLA: An Open-Source Vision-Language-Action Model*. CoRL, 2024.
- [2] M. J. Kim, C. Finn, and P. Liang. *Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success*. 2025.
- [3] A. Rigg, T. Dooms, M. T. Pearce, J. Oramas, and L. Sharkey. *Bilinear MLPs Enable Weight-Based Mechanistic Interpretability*. ICLR, 2025.
- [4] T. Dooms, W. Gauderis, G. A. Wiggins, and J. Oramas. *Compositionality Unlocks Deep Interpretable Models*. 2025.
- [5] F. Oozeer, Y. Erisken, A. Rigg, J. Oramas, and L. Sharkey. *Bilinear Convolution Decomposition for Causal RL Interpretability*. 2024.
- [6] B. Häon, K. Stocking, I. Chuang, and C. Tomlin. *Mechanistic Interpretability for Steering Vision-Language-Action Models*. 2025.
- [7] S. Marks et al. *Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models*. ICLR, 2025.
- [8] M. Kekić et al. *Learning Nonlinear Causal Reductions to Explain Reinforcement Learning Policies*. ICLR, 2026.
- [9] X. Zhou et al. *LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization*. 2025.
- [10] Y. Fang et al. *When Vision Overrides Language: Evaluating and Mitigating Counterfactual Failures in VLAs*. 2026.
- [11] C. Kim et al. *LIBERO-Para: A Diagnostic Benchmark and Metrics for Paraphrase Robustness in VLA Models*. 2026.
- [12] T. Kolda and B. Bader. *Tensor Decompositions and Applications*. SIAM Review, 2009.
- [13] E. Stoudenmire and D. Schwab. *Supervised Learning with Tensor Networks*. NeurIPS, 2016.
- [14] G. Chrysos et al. *Deep Polynomial Neural Networks*. CVPR and TPAMI, 2020 to 2021.

A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in 189/200 trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

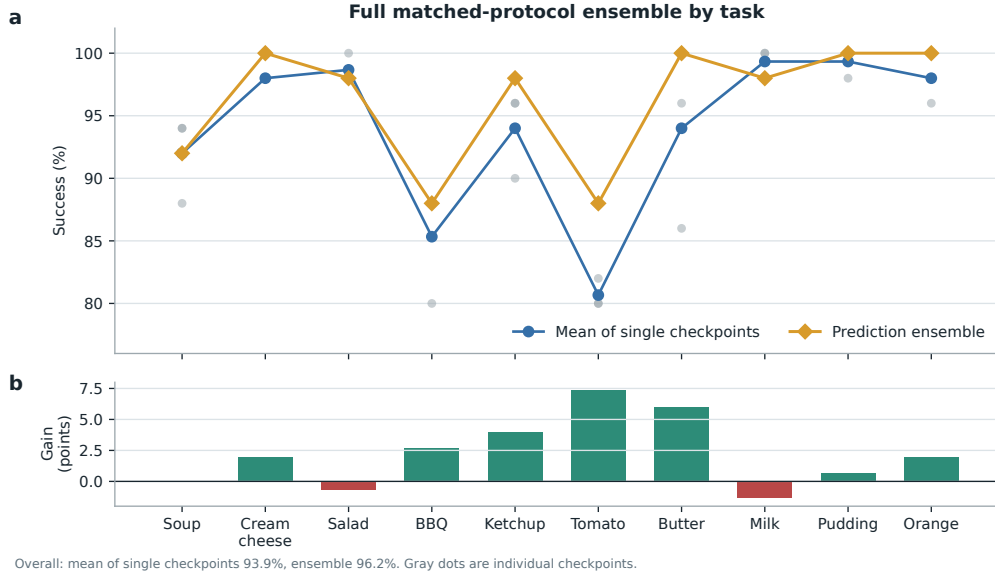


Figure 3: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

230 **A.3 Product-routing multimodal head**

231 For an action-query state h , the product-routing head represents

$$a(h, b) = c_0(h) + \sum_{g=1}^G (2b_g - 1)c_g(h), \quad b \in \{0, 1\}^G. \quad (4)$$

232 The center, offsets, and gates are bilinear CP modules. A final external sign choice selects one of
 233 2^G modes. The head folds to numerical precision, but its five-seed closed-loop mean is $44\% \pm 30\%$
 234 and its range overlaps the linear baseline. The strongest rational policy therefore uses the linear head.
 235 Product routing remains a construction result until it wins a controlled genuinely multimodal task.

236 **A.4 Development interventions**

237 Focused DAgger on BBQ and tomato improves combined success from 82.5% to 96.0% using seven
 238 corrective episodes and 1,131 frames. Applying the same update thinly across all ten tasks reduces
 239 overall success from 92.0% to 72.0%. Three original AWR runs decline by 8.0, 4.6, and 2.6 points.
 240 A reward audit shows that accumulated step cost makes every successful trajectory have negative
 241 return. After removing that term and repairing the terminal boundary, one pilot improves by two
 242 points. These are development diagnostics, not validated paper contributions.

What the July 30 intervention experiments taught us

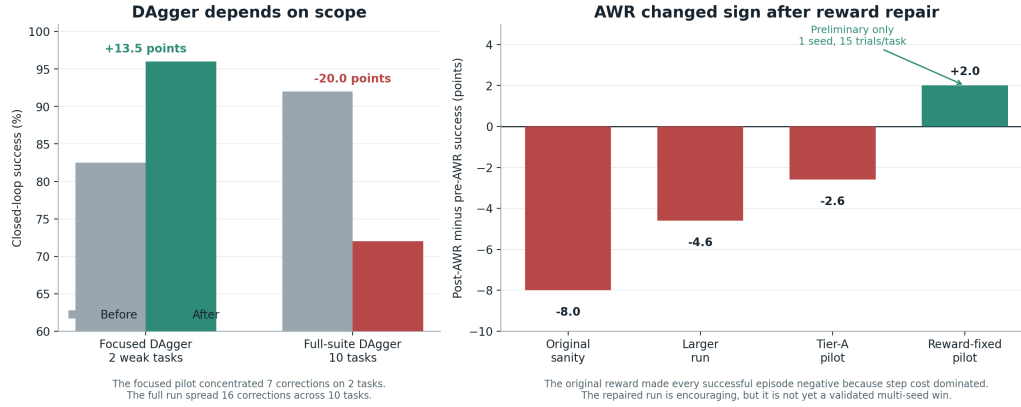


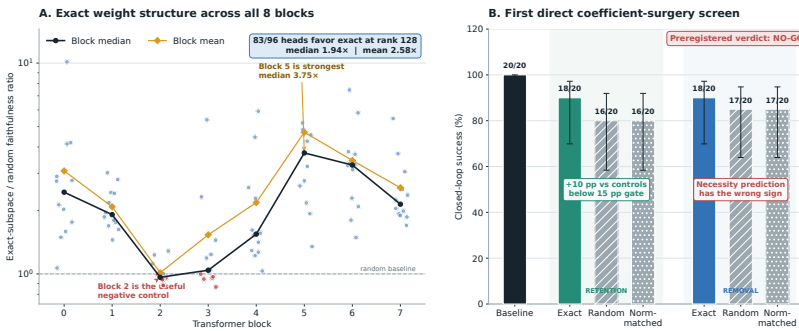
Figure 4: **Development interventions reveal scope and reward sensitivity.** Focused DAGger helps two weak tasks, while spreading a small correction budget across ten tasks causes regression. Three AWR variants also regress before a reward audit and repair. The positive reward-repaired AWR result is a one-seed pilot with 15 trials per task, so it is not evidence of a validated improvement.

243 A.5 Expanded attention screen and first direct edit

244 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 245 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 246 is 2.58, and blocks 2 and 3 provide useful weak or negative controls. At ranks 8 and 32, only 54/96
 247 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at
 248 the wider tested rank.

249 The first direct weight-edit screen uses four tasks and five paired canonical episodes per task. Exact-
 250 subspace retention preserves 18/20 successes compared with 16/20 for both random controls. This
 251 10-point advantage misses the preregistered 15-point gate. Exact removal also preserves 18/20,
 252 compared with 17/20 for both random controls, which is the wrong direction for the necessity
 253 prediction. The screen is a no-go for the current mapping from Gram eigenspaces to query, key, and
 254 value column edits.

Weights expose reproducible structure, but the first direct edit is not yet selective



A: 96 block-head pairs, fixed random-control banks, rank 128. Exact directions come from weights; cached activations only score layer faithfulness. B: 4 tasks × 5 paired canonical episodes. Direct QKV weight edits, no hooks, gradients, retraining, or rollout fitting. Error bars are 95% Wilson intervals.

Figure 5: **Expanded weight-structure and direct-surgery results.** Left: all 96 block-head pairs at rank 128, with fixed random-control banks. Exact directions come from weights, while cached activations only score layer faithfulness. Right: the first direct query-key-value column-edit screen over four tasks and five paired trials per task. Error bars are 95% Wilson intervals. The preregistered verdict is no-go because retention misses the effect-size gate and removal has the wrong sign.

255 A.6 Numerical scope of rational conversion

256 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
 257 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
 258 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
 259 establish practical numerical safety under distribution shift.

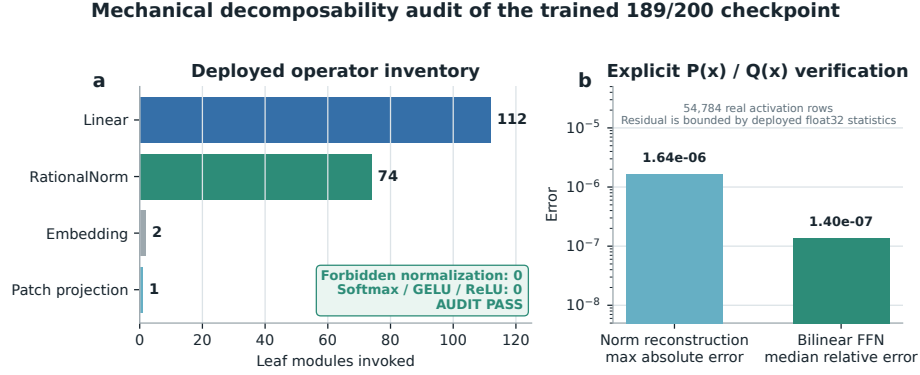


Figure 6: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.